

Full-production discretization P3: scoped T4 tier review

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

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1. Purpose and scope

The claim is promoted from T3 COMPUTABLE-SCAFFOLD to T4 STRONG-EVIDENCE under TSv2. The promotion is deliberately limited to the frozen P3 evidence scope: the fixed 16^3 torus, the hash-pinned P1/P2 canonical functional, $\eta_{\text{shell}} = 0$, the declared grids and manufactured fields, and the one recorded CUDA device/precision matrix. It is not a continuum-error theorem and does not close the gate A3-FULL-DISCRETIZATION-CLOSURE.

1 Why T4 is justified

TSv2 defines T4 as numerics plus analytic bounds plus a partial audit, while explicitly stating that T4 is not a theorem. P3 now has all three components.

1. The spatial residual, real energy-gradient identity, manufactured orders, finite-time trajectories, and Hessian/Ritz tests are executed at their frozen grids.
2. The independent portable functional supplies a second implementation of the energy/residual path, and the CPU/CUDA complex128/complex64 matrix is closed on the recorded RTX 5070 Laptop GPU environment.
3. P2 smoothing gives the positive-time solution-ball rate $C(R, \tau, T)N^{-2}$, and Stage 7 makes its all-time H^2 energy-ball input $M_2(R)$ explicit for the declared radii.
4. The hash-bound integrated verifier checks the immutable evidence matrix at 79/79 PASS.

These facts exceed a mere runnable scaffold. They are strong convergent evidence with analytic control in a pinned scope, which is exactly T4.

2 Why T5 is not justified

The constant $C(R, \tau, T)$ is still unevaluated because the P2 proof uses unnamed norm-equivalence, Moser, Holder, and endpoint-Duhamel constants. A dealiased finite-time Galerkin evolution estimate has also not been propagated from the residual bound. Therefore the package cannot attach a controlled continuum error bar to arbitrary P2 trajectories or historical N32/N64/N128 Sector-B runs. The open gate and T5 blocker remain explicit.

3 Devil's-advocate tier review

1. **Objection: the N8/N12/N16 manufactured-field convergence is a continuum theorem. UPHOLD as false.** It is strong numerical evidence only; the solution-ball constant is not enclosed.

2. **Objection: one CUDA device proves hardware-independent equivalence. VALID WITH MITIGATION.** The claim scope names the exact Torch/CUDA environment and makes no all-device assertion.
3. **Objection: the Hessian/Ritz test proves the full Sector-B spectrum. UPHELD as false.** The evidence covers one homogeneous invariant block and one declared nonuniform fixed Ritz subspace.
4. **Objection: the qualitative N^{-2} rate already closes T5. UPHELD as false.** T5 requires the computable constant, the dealiased finite-time passage, and a reproduction package in the final pinned scope.
5. **Objection: the large $M_2(R)$ values are fitted trajectory outputs. DISMISSED.** They are upper enclosures derived from the Fourier embedding, P1 coefficients, and P2 coercivity; no trajectory fit is used.

4 Next quantitative gate

The next accepted result must expose the actual norm-equivalence, nonlinear Moser, Holder, and semi-group constants needed to turn $M_2(R)$ into an explicit $B_6(R, \tau, T)$. If that majorant is finite but extremely large, it remains a valid enclosure; sharpness is secondary to correctness. Afterward the residual constant must be propagated through a dealiased finite-time evolution estimate before a T5 review.

Result footer

Result ID:	A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:	Strong numerical and analytic discretization evidence in the frozen P3 scope.
Scope:	Fixed T3, canonical P1/P2, eta_shell=0, declared grids/fields and recorded RTX 5070 CUDA matrix.
Dependencies:	A1 production functional; A2 T6 well-posedness.
Evidence grade:	ANALYTIC + EXECUTED + CONDITIONAL.
Reproduction command:	python codes/foundations/ a3_full_production_verify.py --reuse-recorded-audits
Expected output:	79/79; A3-FULL-PRODUCTION-VERIFY-PASS.
Falsification gate:	A3-FULL-DISCRETIZATION-CLOSURE remains open.
Tier before / after:	T3 / T4 STRONG-EVIDENCE.
No-overclaim statement:	Not T5/T6/T7, not a numerical $C(R, \tau, T)$, not a continuum certificate for Sector-B solver runs.
Next required action:	Explicit B6 and C constants, then dealiased evolution.